Università Commerciale Luigi Bocconi

Curie–Weiss & Diffusion

Part II: Reverse OU Dynamics, Collective Closure, and Effective Potentials

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Contents

1 What problem are we actually solving? 2

2 Universal OU structure: forward channel, score, and reverse drift 3

3 Curie–Weiss structure and the emergence of scalar closure 4

4 Closed collective dynamics and the effective potential 5

5 How the potential changes during reverse diffusion 6

6 From two wells to three: a first structural test 7

7 From three wells to n wells: the next closure problem 8

8 Main conclusions 9

Appendices 10

A Universal OU machinery: concise derivation 10
A.1 Forward solution and Gaussian channel . . . 10
A.2 Itô generator and Fokker–Planck equation . . . 10
A.3 Probability current and reverse drift . . . 10
A.4 Exact OU score identity . . . 11
A.5 Linear projections, noise geometry, and the potential criterion . . . 11

B Three magnetization setting: looking for a trend 12
B.1 Clean prior and the Hubbard–Stratonovich variable . . . 12
B.2 Exact $P_0(m)$, coexistence, and three branch weights . . . 13
B.3 Diffuse each branch: exact $P_t(x \mid m)$ and $P_t(m \mid x)$ . . . 14
B.4 High-noise expansion: why the branch evidence becomes scalar . . . 15
B.5 Projection and the candidate three-well potential . . . 16
B.6 Exact projected marginal: check and limitation of closure . . . 17
B.7 A first bridge toward n wells: number of phases versus latent dimension . . . 18

C Curie–Weiss reduction: complete derivation from the prior to the potential 19
C.1 Clean prior, scaling, and Hubbard–Stratonovich decoupling . . . 19
C.2 Marginal magnetization law, phase concentration, and branch masses . . . 19
C.3 Exact branch algebra after OU diffusion . . . 20
C.4 Controlled large-time closure . . . 21
C.5 Projection, exact-in-Δ_t potential, and large-time form . . . 21

Scope and central question

We study the standard Ornstein–Uhlenbeck diffusion of a high-dimensional clean law $P_0(a)$ and ask when its reverse dynamics can be represented by a low-dimensional potential.
Three logically different statements must be separated:

1. The *score* $\nabla_x \log P_t(x)$ is a gradient by definition.
2. For the standard OU process, the complete reverse drift is also an exact gradient in the full N-dimensional state space.
3. The nontrivial problem is whether this high-dimensional dynamics closes on a small set of collective coordinates. Only after closure do we ask whether the resulting reduced force is generated by a scalar potential.

The Curie–Weiss model provides an exceptionally clean example. Its rank-one interaction, one-dimensional latent magnetization, binary-spin algebra, and large-time branch likelihood combine to close the reverse score on one scalar coordinate (Appendix C.4). We then repeat the calculation for three-state spins, which can exhibit three coexisting magnetization phases without requiring a second collective coordinate at the same leading high-noise order (Appendix B.4). The central two-branch chain is

$$\boxed{\begin{aligned} &P_0(a) \longrightarrow P_t(x) \longrightarrow \nabla_x \log P_t(x) \longrightarrow \text{reverse OU drift} \\ &\qquad \longrightarrow m = \pm m^* \longrightarrow M(x) \longrightarrow \mu(t) \longrightarrow V(\mu, t). \end{aligned}}$$

The three-branch comparison leads to the next question: what is required for a potential with n wells, and when does it close on one scalar rather than several coordinates? The relevant distinction is developed in Appendix B.7.

Notation and approximation hierarchy

In the binary Curie–Weiss sections the clean spin vector is $a = (a_1, \dots, a_N)$ with $a_i \in \{\pm 1\}$; the three-state extension allows $a_i \in \{-1, 0, +1\}$ (Appendix B.1). The forward diffused vector is $x(t)$, and

$$\Delta_t := T(1 - e^{-2t}).$$

The thermodynamic auxiliary magnetization is m, while

$$M(x) := \frac{1}{N} \sum_{i=1}^{N} x_i, \qquad \mu(t) := \frac{1}{\sqrt{N}} \sum_{i=1}^{N} x_i(t) = \sqrt{N}\, M(x(t)).$$

We use

$$F_i(x, t) := \partial_{x_i} \log P_t(x)$$

for the score, $\zeta_i := x_i e^{-t}/\Delta_t$ for the local noised field, and $Z(\mu, t)$ for the collective field entering the reduced force.
The OU kernel, reverse drift, exact score identity, and Hubbard–Stratonovich transformation are exact. The two-delta-peak thermodynamic reduction uses $N \to \infty$. Closure on $M(x)$ uses the large-time condition $\max_i |\zeta_i| \ll 1$. The simplest displayed potential additionally uses $\Delta_t \simeq T$ (Appendix C.5). The three-branch extension has its own thermodynamic coexistence and high-noise assumptions (Appendix B.2; Appendix B.4).

1 What problem are we actually solving?

Objective. We first isolate the genuinely nontrivial question. This avoids confusing three different uses of the word "potential."

Object	Meaning	Status / appendix
$x_t \mid a \sim \mathcal{N}(e^{-t}a, \Delta_t I)$	Forward OU channel	exact (A.1)
$b_{\mathrm{R}} = -b + 2T\nabla \log P_t$	Reverse drift at constant diffusion	exact (A.3)
$F_i = -(x_i - e^{-t}\mathbb{E}[a_i \mid x])/\Delta_t$	OU score identity	exact (A.4)
$P_0(m,a)$: HS augmentation	Latent-variable representation	exact (C.1)
$P_0(m) \Rightarrow W_+\delta_{m^*} + W_-\delta_{-m^*}$	Two-state thermodynamic reduction	large N (C.2)
$\mathbb{E}[a_i \mid x] = \mathcal{G}(M(x), t)$	Scalar closure	large t (C.4)
$V(\mu, t)$ with $\Delta_t = T$	Simple effective potential	large N, large t (C.5)
Three-state $P_0(m)$ and $V_3(\mu, t)$	Three competing magnetization phases	exact $P_0(m)$; asymptotic V_3 (B.2; B.5)

Table 1: Exact identities, approximation levels, and clickable appendix derivations.

At every time t for which P_t is smooth and strictly positive, the score is

$$F(x,t) = \nabla_x \log P_t(x).$$

It is therefore already a gradient field. This fact alone does not produce a useful low-dimensional description (Appendix A.4).

For the standard OU process, the reverse deterministic drift in an increasing reverse clock is

$$B(x,t) = x + 2T\nabla_x \log P_t(x).$$

Consequently

$$B(x,t) = -\nabla_x U_t(x), \qquad U_t(x) := -\frac{1}{2}\|x\|^2 - 2T \log P_t(x). \tag{1}$$

up to an additive function of time. Thus an exact time-dependent potential already exists in the full N-dimensional space (Appendix A.3).

The research problem is stronger:

$$\boxed{\text{Find a small collection } q(x) \text{ such that the projected reverse dynamics closes on } q.} \tag{2}$$

If the reduced state has one coordinate, a scalar potential can always be obtained locally by integration. For several reduced coordinates, the existence of one scalar potential becomes a nontrivial integrability condition (Appendix A.5).

The central distinction

The score being a gradient is universal. The Curie–Weiss achievement is that the model-specific posterior mean $\mathbb{E}[a_i \mid x]$ becomes a function of one scalar statistic $M(x)$, identical for every coordinate. That closure, not the final integration, is the main analytical result (Appendix C.4; Appendix C.5).

2 Universal OU structure: forward channel, score, and reverse drift

Objective. We derive the part of the argument that depends only on the linear Gaussian OU channel and is therefore available for any clean law P_0.

The forward Itô process is

$$\mathrm{d}x_t = -x_t\,\mathrm{d}t + \sqrt{2T}\,\mathrm{d}W_t, \qquad x(0) = a. \tag{3}$$

Its explicit solution gives

$$\boxed{x(t) \mid a \sim \mathcal{N}(e^{-t}a, \Delta_t I), \qquad \Delta_t = T(1 - e^{-2t}).} \tag{4}$$

Hence

$$P_t(x) = \int \mathrm{d}a\, P_0(a) \frac{1}{(2\pi\Delta_t)^{N/2}} \exp\left[-\frac{\|x - e^{-t}a\|^2}{2\Delta_t}\right]. \tag{5}$$

The OU kernel and convolution follow directly from the explicit solution (Appendix A.1).

Differentiating this convolution gives the exact score identity

$$\boxed{F_i(x,t) = \partial_{x_i} \log P_t(x) = -\frac{x_i - e^{-t}\mathbb{E}[a_i \mid x]}{\Delta_t}.} \tag{6}$$

All dependence on the clean law is concentrated in the denoiser $\mathbb{E}[a_i \mid x]$ (Appendix A.4).

To state the reverse dynamics without a sign ambiguity, introduce an increasing reverse clock

$$s := t_f - t.$$

Then the reverse process is

$$\boxed{\mathrm{d}y_s = \left[y_s + 2T\nabla_y \log P_{t_f - s}(y_s)\right]\, \mathrm{d}s + \sqrt{2T}\, \mathrm{d}\overline{W}_s.} \tag{7}$$

In the decreasing forward-time convention this is

$$\boxed{-\frac{\mathrm{d}y_i}{\mathrm{d}t} = y_i + 2TF_i(y,t) + \eta_i(t).} \tag{8}$$

The connection from Itô calculus to the Fokker–Planck equation is derived in Appendix A.2; the time-reversal argument and its factor $2T$ are derived in Appendix A.3.

3 Curie–Weiss structure and the emergence of scalar closure

Objective. We now identify the special structural ingredients that make the generic OU score in (6) close on a single collective coordinate.

The clean law is

$$P_0(a) \propto \exp\left[\frac{\beta}{2N}\left(\sum_{i=1}^N a_i\right)^2 + \frac{h}{N}\sum_{i=1}^N a_i\right], \qquad a_i \in \{\pm 1\}. \tag{9}$$

The interaction depends only on the rank-one statistic $\sum_i a_i$. A one-dimensional Hubbard–Stratonovich variable therefore linearizes the quadratic term, yielding

$$P_0(m,a) \propto \exp\left(-\frac{\beta N}{2}m^2\right) \prod_{i=1}^N \exp[(\beta m + h/N)a_i]\,. \tag{10}$$

Conditional on m, the spins are independent (Appendix C.1). Summing them out gives

$$P_0(m) \asymp e^{-\beta N \Phi_N(m)}, \qquad \Phi_N(m) = \frac{m^2}{2} - \frac{1}{\beta}\log\cosh(\beta m + h/N). \tag{11}$$

For $\beta > 1$, Laplace concentration produces two stable branches near $\pm m^*$, where

$$m^* = \tanh(\beta m^*). \tag{12}$$

Their normalized masses are

$$W_\pm = \frac{e^{\pm h m^*}}{2\cosh(h m^*)}. \tag{13}$$

Thus, at leading order in N, the thermodynamic latent variable becomes a two-state label (Appendix C.2).

Conditional on branch $\pm$, the OU-diffused branch likelihood contains the branch-dependent factor

$$Q_\pm(x) = \prod_{j=1}^N \left[1 \pm m^* \tanh \zeta_j\right], \qquad \zeta_j := \frac{x_j e^{-t}}{\Delta_t}. \tag{14}$$

The exact branch posterior within the two-state approximation is

$$\boxed{P(\pm \mid x) = \frac{W_{\pm} Q_{\pm}(x)}{W_{+} Q_{+}(x) + W_{-} Q_{-}(x)}.} \tag{15}$$

This formula displays the prior branch mass and the diffused branch likelihood separately (Appendix C.3).

The cleanest large-time reduction is obtained from the likelihood ratio, not from the two absolute likelihoods separately:

$$\begin{aligned} \log \frac{Q_{+}(x)}{Q_{-}(x)} &= \sum_{j=1}^{N} \log \frac{1 + m^{*} \tanh \zeta_j}{1 - m^{*} \tanh \zeta_j} \\ &= 2 \sum_{j=1}^{N} \operatorname{artanh}(m^{*} \tanh \zeta_j) \\ &= 2m^{*} \sum_{j=1}^{N} \zeta_j + O\left(\sum_{j=1}^{N} |\zeta_j|^3 \right). \end{aligned} \tag{16}$$

There is no quadratic term in the ratio; the sign-independent quadratic contribution cancels exactly. Since

$$\sum_j \zeta_j = \frac{e^{-t}}{\Delta_t} \sum_j x_j = N M(x) \frac{e^{-t}}{\Delta_t},$$

the branch posterior depends, at leading order, only on $M(x)$ (Appendix C.4).

At the same order, the branch-conditional local means become $\pm m^{*}$. Combining them with (15) gives

$$\boxed{\mathbb{E}[a_i \mid x] \simeq m^{*} \tanh\left[m^{*} \left(h + N M(x) \frac{e^{-t}}{\Delta_t} \right) \right].} \tag{17}$$

The right-hand side is independent of the site label i. Substitution into (6) gives the closed large-time score

$$\boxed{F_i^{\mathrm{LT}}(x,t) = -\frac{x_i}{\Delta_t} + m^{*} \frac{e^{-t}}{\Delta_t} \tanh\left[m^{*} \left(h + N M(x) \frac{e^{-t}}{\Delta_t} \right) \right].} \tag{18}$$

The peak masses come from Appendix C.2, the exact branch algebra from Appendix C.3, and the controlled high-noise approximation from Appendix C.4.

Why the dimensional reduction works

The interaction is rank one, so one auxiliary magnetization is sufficient. Permutation symmetry makes every coordinate couple to that same latent variable. At large time, the branch log-likelihood ratio accumulates the weak local evidence only through $\sum_i x_i$. These three facts turn the model-specific denoiser into the same scalar function of $M(x)$ at every site (Appendix C.1; Appendix C.4).

4 Closed collective dynamics and the effective potential

Objective. We project the closed score onto the collective coordinate and distinguish the existence of a potential from the additional approximation $\Delta_t \simeq T$.

Define

$$\boxed{\mu(t) := \frac{1}{\sqrt{N}} \sum_{i=1}^{N} x_i(t) = \sqrt{N}\, M(x(t)).} \tag{19}$$

The projected noise

$$\eta_{\mu}(t) := \frac{1}{\sqrt{N}} \sum_i \eta_i(t)$$

satisfies

$$\langle \eta_\mu(t)\eta_\mu(t')\rangle = 2T\delta(t-t'). \tag{20}$$

The noise calculation is given in Appendix C.5; its linear projection mechanism is also recorded in Appendix A.5. Summing the reverse equations and inserting (18) gives, before approximating Δ_t,

$$-\frac{\mathrm{d}\mu}{\mathrm{d}t} = \left(1-\frac{2T}{\Delta_t}\right)\mu + 2Tm^*\sqrt{N}\frac{e^{-t}}{\Delta_t}\tanh\left[m^*\left(h+\sqrt{N}\mu\frac{e^{-t}}{\Delta_t}\right)\right] + \eta_\mu(t). \tag{21}$$

This one-dimensional force already has a potential. Define

$$\boxed{V_\Delta(\mu,t) = \frac{1}{2}\left(\frac{2T}{\Delta_t}-1\right)\mu^2 - 2T\log\cosh\left[m^*\left(h+\frac{\sqrt{N}e^{-t}}{\Delta_t}\mu\right)\right].} \tag{22}$$

Then

$$-\frac{\mathrm{d}\mu}{\mathrm{d}t} = -\partial_\mu V_\Delta(\mu,t) + \eta_\mu(t). \tag{23}$$

Thus $\Delta_t \simeq T$ is not needed for the existence of a potential. It is only needed for the simpler form used near the high-noise transition (Appendix C.5).

At large forward time,

$$\Delta_t = T(1-e^{-2t}) \simeq T,$$

so

$$\boxed{-\frac{\mathrm{d}\mu}{\mathrm{d}t} = -\mu + 2m^*\sqrt{N}e^{-t}\tanh Z(\mu,t) + \eta_\mu(t),} \tag{24}$$

where

$$Z(\mu,t) := m^*\left(h+\frac{\sqrt{N}}{T}e^{-t}\mu\right). \tag{25}$$

Using $(\log\cosh Z)' = \tanh Z$, the simplified potential is

$$\boxed{V(\mu,t) = \frac{1}{2}\mu^2 - 2T\log\cosh Z(\mu,t).} \tag{26}$$

The reduction to this expression is verified by differentiation in Appendix C.5.

For stability it is preferable to return to the increasing reverse clock $s = t_f - t$. The deterministic reduced equation is then

$$\frac{\mathrm{d}\mu}{\mathrm{d}s} = -\partial_\mu V(\mu, t_f - s). \tag{27}$$

Therefore a stationary point is locally stable precisely when it is a local minimum of V, equivalently when $V'' > 0$ (Appendix C.5).

5 How the potential changes during reverse diffusion

Objective. We interpret the time-dependent potential and locate the loss of stability of the uninformative state.

In the symmetric case $h = 0$, the origin is stationary. Expanding $\log\cosh Z = Z^2/2 + O(Z^4)$ gives

$$V(\mu,t) = \frac{1}{2}\left[1-\frac{2m^{*2}N}{T}e^{-2t}\right]\mu^2 + O(\mu^4). \tag{28}$$

Hence

$$V''(0,t) = 1 - \frac{2m^{*2}N}{T}e^{-2t}. \tag{29}$$

The origin loses stability when

$$e^{-2t} = \frac{T}{2m^{*2}N}, \qquad \text{so} \qquad \boxed{t_{\mathrm{spec}} = \frac{1}{2}\log\left(\frac{2m^{*2}N}{T}\right).} \tag{30}$$

These curvature and speciation expressions are evaluated in Appendix C.5. In particular,

$$t_{\mathrm{spec}} = \frac{1}{2}\log N + O(1).$$

Later in reverse generation, t is smaller and the collective argument $|Z|$ becomes large. Since

$$\log\cosh Z = |Z| - \log 2 + o(1),$$

the symmetric potential becomes, up to μ-independent terms,

$$V(\mu, t) \simeq \frac{1}{2}\mu^2 - 2m^*\sqrt{N}e^{-t}|\mu|. \tag{31}$$

The origin has become unstable and two ferromagnetic wells have formed (Appendix C.5).

The local expansion is self-consistent at the transition. For typical high-noise coordinates,

$$\max_i |x_i| = O_{\mathbb{P}}(\sqrt{T\log N}),$$

while $e^{-t_{\mathrm{spec}}} = O(N^{-1/2})$. Thus

$$\max_i |\zeta_i| = O_{\mathbb{P}}\left(\sqrt{\frac{\log N}{N}}\right) \longrightarrow 0. \tag{32}$$

At the same time,

$$\Delta_{t_{\mathrm{spec}}} = T\left(1 - O(N^{-1})\right),$$

which justifies $\Delta_t \simeq T$ at the transition scale (Appendix C.4; Appendix C.5).

6 From two wells to three: a first structural test

Objective. We summarize the three-magnetization extension before asking what changes for an arbitrary number of wells. The full calculation follows the same steps as the binary model in Appendix B.

Allow $a_i \in \{-1, 0, +1\}$ and add an occupancy penalty to the Curie–Weiss interaction:

$$P_0(a) \propto \exp\left[\frac{\beta}{2N}\left(\sum_i a_i\right)^2 - \lambda\sum_i a_i^2\right]. \tag{33}$$

The same one-scalar Hubbard–Stratonovich step gives an exact marginal $P_0(m) \propto e^{-\beta N\Phi_\lambda(m)}$, with

$$\Phi_\lambda(m) = \frac{m^2}{2} - \frac{1}{\beta}\log\left[1 + 2e^{-\lambda}\cosh(\beta m)\right]. \tag{34}$$

Unlike the Ising case, suitable coexistence parameters produce three thermodynamic minima $-m^*$, 0, and $+m^*$. For instance, at $\beta = 4$ one finds $\lambda \simeq 1.90506$ and $m^* \simeq 0.73148$. Equal free-energy depths ensure that all three phases survive at large N, but their masses need not be 1/3: the Laplace curvatures determine the weights $W_- = W_+ = W$ and W_0 (Appendix B.1; Appendix B.2).

Passing the three branches through the same OU channel gives an exact one-dimensional integral for $P_t(m \mid x)$, which still depends on the individual x_i (Appendix B.3). At the high-noise scale $t = \frac{1}{2}\log N + O(1)$, however, the branch evidence depends to leading order only on $\mu = N^{-1/2}\sum_i x_i$; the different branch variances cancel from the leading likelihood ratio after concentration (Appendix B.4). With $c_t = m^*\sqrt{N}e^{-t}$ and $\Delta_t \simeq T$, the resulting candidate scalar reverse potential is

$$\boxed{V_3(\mu, t) \simeq \frac{1}{2}\mu^2 - 2T\log\left[W_0 + 2We^{-c_t^2/(2T)}\cosh\left(\frac{c_t\mu}{T}\right)\right] + C(t).} \tag{35}$$

The central branch contributes an extra term inside the logarithm; three clean phases do not guarantee three visible wells at every reverse time. This is a leading-order *full-score closure* result, distinct from the exact potential of the projected one-dimensional OU marginal (Appendix B.5; Appendix B.6).

Emerging trend. The two-branch and three-branch models both close on the *same one* collective coordinate at the stated high-noise order. Hence the number of thermodynamic phases does not, by itself, fix the number of latent coordinates. The question for n wells is whether branch evidence and within-branch denoising can still be represented by a small set of collective statistics (Appendix B.7).

7 From three wells to n wells: the next closure problem

Objective. We now state the next research problem: how does the smallest closing representation change when the number of phases grows, possibly with several independent directions? The Gaussian branch-geometry example in Appendix B.7 clarifies why the number of wells and the latent dimension need not coincide. The projection and potential conditions are spelled out in Appendix A.5.

Let

$$h^1, \dots, h^p \in \mathbb{R}^D, \qquad \|h^\alpha\|_2 = 1,$$

with no initial assumption on their overlaps. A genuine latent-template model is

$$P_0(\alpha) = w_\alpha, \qquad P_0(a \mid \alpha) = \frac{1}{Z_\alpha} \exp(\beta \langle a, h^\alpha \rangle), \tag{36}$$

so that

$$\boxed{P_0(a) = \sum_{\alpha=1}^{p} w_\alpha P_0(a \mid \alpha) \propto \sum_{\alpha=1}^{p} \widetilde{w}_\alpha e^{\beta \langle a, h^\alpha \rangle}.} \tag{37}$$

This must be distinguished from

$$\exp\left(\beta \sum_{\alpha=1}^{p} \langle a, h^\alpha \rangle\right) = \exp\left(\beta \left\langle a, \sum_{\alpha} h^\alpha \right\rangle\right),$$

which is only one effective linear field and does not represent competing latent templates (Appendix B.7).

The universal OU identities remain unchanged:

$$F_i(x,t) = -\frac{x_i - e^{-t}\mathbb{E}[a_i \mid x]}{\Delta_t}, \qquad b_{\mathrm{R}} = x + 2T\nabla \log P_t.$$

The model-specific question is whether $\mathbb{E}[a \mid x]$ depends on x only through finitely many overlaps, naturally (Appendix B.7)

$$q_\alpha(x) := \langle h^\alpha, x \rangle, \qquad \alpha = 1, \dots, p. \tag{38}$$

A useful closure would have the form

$$\mathbb{E}[a \mid x] = \mathcal{D}(q_1(x), \dots, q_p(x), t) + \text{controlled remainder}. \tag{39}$$

Projecting isotropic OU noise onto these coordinates gives correlated noise:

$$\langle \eta_\alpha(t) \eta_\beta(t') \rangle = 2T G_{\alpha\beta} \delta(t - t'), \qquad G_{\alpha\beta} := \langle h^\alpha, h^\beta \rangle. \tag{40}$$

Thus the Gram matrix is unavoidable even before any additional structure is assumed (Appendix A.5).

If the closed reduced drift is $\mathcal{F}(q,t)$, an ordinary Euclidean potential

$$\mathcal{F} = -\nabla_q V$$

requires, on a simply connected domain,

$$\boxed{\partial_{q_\beta} \mathcal{F}_\alpha = \partial_{q_\alpha} \mathcal{F}_\beta \qquad \text{for all } \alpha, \beta.} \tag{41}$$

If the natural mobility is the Gram matrix and $\mathcal{F} = -G\nabla_q V$, then the corresponding condition is that $G^{-1}\mathcal{F}$ be curl free on the nondegenerate template span (Appendix A.5).

Research program suggested by the Curie–Weiss example

The next investigation should proceed in this order:

1. derive the exact posterior/score for the p-template mixture;
2. identify the smallest set of overlaps on which it closes, exactly or asymptotically (Appendix B.7);
3. derive the projected noise covariance and drift, retaining the Gram geometry (Appendix A.5);
4. test whether the reduced force is conservative in the appropriate metric (Appendix A.5);
5. only then study the shape, bifurcations, and possible hierarchical wells of the resulting potential.

8 Main conclusions

Objective. We connect the two- and three-well results and formulate the problem that remains for an arbitrary number of wells and directions.

The full OU reverse drift has an exact potential in the original state space, and the score is always a gradient (Appendix A.3; Appendix A.4). Neither fact explains the special Curie–Weiss reduction: in the symmetric binary model, two macroscopic branches lead to a denoiser depending on the mean $M(x)$ alone, and therefore to a one-dimensional double-well potential (Appendix C.2; Appendix C.4; Appendix C.5).

Changing the clean law to the three-state model yields three coexisting magnetization phases but preserves scalar closure at the same leading high-noise order, giving the candidate potential (35) (Appendix B.2; Appendix B.4; Appendix B.5). This is the trend motivating our next question: for n wells, which assumptions keep the relevant information one-dimensional, and which demand several independent collective coordinates? The difference between branch sufficiency and actual reverse-score closure is explained in Appendix B.7; the additional metric and integrability issues for multi-directional models are derived in Appendix A.5.

Appendices

Appendix A records the universal OU derivation concisely. Appendix B follows the same reasoning for three-state spins and compares exact results with their high-noise closure. Appendix C retains the continuous, detailed Ising Curie–Weiss derivation from the clean prior to the potential.

A Universal OU machinery: concise derivation

Objective. This appendix supplies the universal stochastic-calculus steps used in the main text without interrupting the Curie–Weiss argument.

A.1 Forward solution and Gaussian channel

For one coordinate,

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} + x_i = \eta_i(t), \qquad x_i(0) = a_i.$$

Multiplication by e^t and integration give

$$x_i(t) = e^{-t}a_i + \int_0^t e^{-(t-s)}\eta_i(s)\ \mathrm{d}s.$$

With

$$\langle \eta_i(s)\eta_j(s')\rangle = 2T\delta_{ij}\delta(s-s'),$$

the accumulated-noise variance is

$$2T\int_0^t e^{-2(t-s)}\ \mathrm{d}s = T(1-e^{-2t}) = \Delta_t.$$

This proves (4) and (5).

A.2 Itô generator and Fokker–Planck equation

For

$$\mathrm{d}X_i = b_i(X,t)\ \mathrm{d}t + \sum_k \sigma_{ik}(X,t)\ \mathrm{d}W_k,$$

Itô's formula gives the generator

$$\mathcal{L}f = \sum_i b_i\partial_i f + \frac{1}{2}\sum_{i,j} a_{ij}\partial_i\partial_j f, \qquad a = \sigma\sigma^\top.$$

Transferring derivatives from a test function to the density gives

$$\partial_t P = -\sum_i \partial_i(b_i P) + \frac{1}{2}\sum_{i,j}\partial_i\partial_j(a_{ij}P). \tag{42}$$

For $b = -x$ and $a = 2TI$,

$$\partial_t P_t = \nabla\cdot(xP_t) + T\nabla^2 P_t. \tag{43}$$

A.3 Probability current and reverse drift

For constant isotropic diffusion,

$$\partial_t P = -\nabla\cdot J, \qquad J = bP - T\nabla P.$$

Define $s = t_f - t$ and $Q(y,s) = P_{t_f-s}(y)$. The required reversed density evolution is

$$\partial_s Q = -\nabla\cdot(yQ) - T\nabla^2 Q.$$

A reverse SDE with drift B has

$$\partial_s Q = -\nabla \cdot (BQ) + T\nabla^2 Q.$$

Matching the local currents, with the usual decay/boundary conditions and no additional solenoidal current, gives

$$BQ = yQ + 2T\nabla Q,$$

hence

$$B = y + 2T\nabla \log Q.$$

This proves (7) and (8). The factor $2T$ appears because the diffusive current $-T\nabla P$ must become $+T\nabla P$ when the probability flow is reversed. In particular, the full-space OU reverse drift has the exact gradient representation

$$-\nabla_x \left[-\frac{1}{2}\|x\|^2 - 2T \log P_t(x) \right] = x + 2T\nabla_x \log P_t(x).$$

A.4 Exact OU score identity

Differentiate (5):

$$\partial_{x_i} P_t(x) = -\frac{x_i}{\Delta_t} P_t(x) + \frac{e^{-t}}{\Delta_t} \int \mathrm{d}a \, a_i P_t(a, x). \tag{44}$$

After division by $P_t(x)$,

$$\partial_{x_i} \log P_t(x) = -\frac{x_i}{\Delta_t} + \frac{e^{-t}}{\Delta_t} \int \mathrm{d}a \, a_i P_t(a \mid x),$$

which is (6).

A.5 Linear projections, noise geometry, and the potential criterion

For a matrix $H \in \mathbb{R}^{p \times D}$, define $q = Hx$ for a D-dimensional OU process $\mathrm{d}x = -x \, \mathrm{d}t + \sqrt{2T} \, \mathrm{d}W$. Linearity of the Itô differential gives

$$\mathrm{d}q = -q \, \mathrm{d}t + \sqrt{2T} \, H \, \mathrm{d}W, \qquad \mathrm{d}\langle q_\alpha, q_\beta \rangle_t = 2T(HH^\top)_{\alpha\beta} \, \mathrm{d}t.$$

If the rows of H are the template vectors $(h^\alpha)^\top$, then $G = HH^\top$ is their Gram matrix, independently of whether these directions are orthogonal. For any smooth scalar $V(q)$, the chain rule yields

$$H\nabla_x [V(Hx)] = HH^\top \nabla_q V = G\nabla_q V.$$

Thus projection of an ambient gradient component that factors through q naturally produces a *metric gradient* $-G\nabla_q V$, not necessarily a Euclidean gradient $-\nabla_q V$. This statement presupposes that the rest of the ambient drift also closes on q. For a smooth closed force $\mathcal{F}(q)$ on a simply connected nondegenerate domain, a Euclidean potential exists if and only if $\partial_{q_\beta} \mathcal{F}_\alpha = \partial_{q_\alpha} \mathcal{F}_\beta$. If G is constant and invertible, a metric potential instead exists if and only if $G^{-1}\mathcal{F}$ has symmetric cross-derivatives. For singular G, the statement must be restricted to the independent template span.

B Three magnetization setting: looking for a trend

Objective. We repeat the sequence of the binary Curie–Weiss derivation, but allow $a_i \in \{-1, 0, +1\}$. The central questions are whether the clean measure can support three competing magnetizations, whether their OU-diffused likelihoods can be evaluated, and whether the reverse force still depends only on $\mu = N^{-1/2}\sum_i x_i$. We will distinguish exact finite-N identities from the three-branch and high-noise approximations. The original two-magnetization calculation is developed in full in Appendix C.

B.1 Clean prior and the Hubbard–Stratonovich variable

We begin with the mean-field three-state (Blume–Capel-type) law

$$\boxed{P_0(a) = \frac{1}{\mathcal{Z}_N}\exp\left[\frac{\beta}{2N}\left(\sum_{i=1}^N a_i\right)^2 + \frac{h}{N}\sum_{i=1}^N a_i - \lambda\sum_{i=1}^N a_i^2\right], \qquad a_i \in \{-1, 0, +1\}.} \tag{45}$$

The quadratic interaction is identical to Curie–Weiss. The new parameter λ controls the occupancy of the nonzero states, since $a_i^2 = 1$ for $a_i = \pm 1$ and $a_i^2 = 0$ for $a_i = 0$. Set

$$S(a) := \sum_i a_i, \qquad Q(a) := \sum_i a_i^2.$$

Here S records the magnetization while Q counts occupied (± 1) sites. The new term does not remove the rank-one interaction; thus one scalar Hubbard–Stratonovich variable still suffices. Using the exact Gaussian identity (85) with $S = S(a)$,

$$e^{\beta S^2/(2N)} = \sqrt{\frac{\beta N}{2\pi}}\int_{\mathbb{R}} \mathrm{d}m\, \exp\left[-\frac{\beta N}{2}m^2 + \beta m S\right]. \tag{46}$$

Introduce the shorthand $u_m := \beta m + h/N$. Inserting the identity into (45) gives the augmented joint law

$$P_0(m, a) = \frac{1}{\widetilde{\mathcal{Z}}_N} e^{-\beta N m^2/2}\prod_{i=1}^N e^{u_m a_i - \lambda a_i^2}. \tag{47}$$

Integrating out m returns the original distribution of a. Conditional on m the coordinates become independent, with

$$\boxed{P_0(a \mid m) = \prod_{i=1}^N \frac{e^{u_m a_i - \lambda a_i^2}}{D_\lambda(u_m)}, \qquad D_\lambda(u) := 1 + 2e^{-\lambda}\cosh u.} \tag{48}$$

Indeed, the one-site normalization is the elementary sum

$$\sum_{a_i=-1,0,+1} e^{u a_i - \lambda a_i^2} = e^{-u-\lambda} + 1 + e^{u-\lambda} = 1 + 2e^{-\lambda}\cosh u. \tag{49}$$

It follows, still exactly at finite N, that

$$\begin{aligned} P_0(a_i = 0 \mid m) &= \frac{1}{D_\lambda(u_m)}, \\ P_0(a_i = \pm 1 \mid m) &= \frac{e^{-\lambda \pm u_m}}{D_\lambda(u_m)}. \end{aligned} \tag{50}$$

This replaces the Ising single-site partition function $2\cosh u$ by $1 + 2e^{-\lambda}\cosh u$. No other change was required to perform the decoupling.

B.2 Exact $P_0(m)$, coexistence, and three branch weights

Now sum (47) over a. Because the sites factorize,

$$\begin{aligned} P_0(m) &\propto e^{-\beta N m^2/2} \prod_{i=1}^{N} \sum_{a_i=-1,0,+1} e^{u_m a_i - \lambda a_i^2} \\ &= e^{-\beta N m^2/2} [D_\lambda(u_m)]^N. \end{aligned} \tag{51}$$

Consequently the normalized marginal of the *continuous auxiliary* variable m is

$$\boxed{P_0(m) = \frac{e^{-\beta N m^2/2}[1 + 2e^{-\lambda}\cosh(\beta m + h/N)]^N}{\displaystyle\int_{\mathbb{R}} \mathrm{d}v\, e^{-\beta N v^2/2}[1 + 2e^{-\lambda}\cosh(\beta v + h/N)]^N}.} \tag{52}$$

As before, write $P_0(m) \propto e^{-\beta N \Phi_{\lambda,N}(m)}$, with

$$\boxed{\Phi_{\lambda,N}(m) = \frac{m^2}{2} - \frac{1}{\beta}\log[1 + 2e^{-\lambda}\cosh(\beta m + h/N)].} \tag{53}$$

Thus $P_0(m)$ is known as an exact one-dimensional expression, even before any saddle-point approximation.

The derivative is

$$\Phi'_{\lambda,N}(m) = m - \frac{2e^{-\lambda}\sinh(\beta m + h/N)}{1 + 2e^{-\lambda}\cosh(\beta m + h/N)}. \tag{54}$$

At $h = 0$, a stationary magnetization m_α therefore solves

$$\boxed{m_\alpha = \frac{2e^{-\lambda}\sinh(\beta m_\alpha)}{1 + 2e^{-\lambda}\cosh(\beta m_\alpha)}.} \tag{55}$$

The right-hand side is also $\mathbb{E}[a_i \mid m_\alpha]$ by (50), so a stationary m equals the conditional mean spin. The central solution $m_0 = 0$ always exists; for appropriate (β, λ) there are additional stable solutions $m_\pm = \pm m^*$.

To have three *coexisting macroscopic phases*, mere stationarity is insufficient: all three points must be local minima and have the same leading free-energy value. At $h = 0$, symmetry equates the outer two, so the coexistence conditions reduce to

$$\begin{cases} m^* = \dfrac{2e^{-\lambda}\sinh(\beta m^*)}{1 + 2e^{-\lambda}\cosh(\beta m^*)}, \\ \dfrac{\beta (m^*)^2}{2} = \log\left[\dfrac{1 + 2e^{-\lambda}\cosh(\beta m^*)}{1 + 2e^{-\lambda}}\right], \end{cases} \tag{56}$$

together with $\Phi''_\lambda(0) > 0$ and $\Phi''_\lambda(m^*) > 0$. The second equation is just $\Phi_\lambda(m^*) = \Phi_\lambda(0)$ written out explicitly. For example, solving (56) at $\beta = 4$ gives

$$\lambda_{\mathrm{coex}} \simeq 1.90506335, \qquad m^* \simeq 0.73148205.$$

This is a concrete three-phase setting, rather than an arbitrary three-component mixture imposed by hand.

To calculate the masses of the peaks, expand Φ quadratically around each minimum. Laplace's method gives

$$I_\alpha \simeq e^{-\beta N \Phi(m_\alpha)} \sqrt{\frac{2\pi}{\beta N \Phi''(m_\alpha)}}, \qquad \alpha \in \{-, 0, +\}. \tag{57}$$

At coexistence the exponential factors agree, but the curvatures need not agree. Hence, for $h = 0$,

$$\boxed{W_\alpha = \frac{[\Phi''(m_\alpha)]^{-1/2}}{\displaystyle\sum_{\gamma\in\{-,0,+\}} [\Phi''(m_\gamma)]^{-1/2}} + o(1), \quad W_- = W_+.} \tag{58}$$

For $\beta = 4$ at coexistence, the weights are approximately

$$W_- \simeq 0.282, \qquad W_0 \simeq 0.436, \qquad W_+ \simeq 0.282.$$

Equal free-energy depths do *not* imply three exactly equal masses. Unlike the two Ising branches, the middle peak is not the mirror image of an outer peak. A finite-size adjustment of λ of order N^{-1} can balance the asymptotic masses, but no such adjustment is needed to keep all three phases present.

The resulting thermodynamic approximation is

$$\boxed{P_0(m) \simeq W_-\delta(m+m^*) + W_0\delta(m) + W_+\delta(m-m^*).} \tag{59}$$

The notation m here refers to the Hubbard–Stratonovich variable. The actual discrete magnetization $S(a)/N$ is a different finite-N random variable, though both concentrate at these same phase values.

B.3 Diffuse each branch: exact $P_t(x \mid m)$ and $P_t(m \mid x)$

Pass the clean spins through the same OU channel as in Appendix A:

$$P_t(x \mid a) = \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi\Delta_t}} \exp\left[-\frac{(x_i - e^{-t}a_i)^2}{2\Delta_t}\right], \quad \Delta_t = T(1-e^{-2t}). \tag{60}$$

We wish to calculate $P_t(x \mid m) = \sum_a P_t(x \mid a)P_0(a \mid m)$. Write

$$\zeta_i := \frac{e^{-t}x_i}{\Delta_t}, \qquad \kappa_t := \frac{e^{-2t}}{2\Delta_t}.$$

Expanding the Gaussian square reveals the key difference from Ising spins:

$$-\frac{(x_i - e^{-t}a_i)^2}{2\Delta_t} = -\frac{x_i^2}{2\Delta_t} + \zeta_i a_i - \kappa_t a_i^2. \tag{61}$$

For $a_i = \pm 1$ only, $a_i^2 = 1$ is constant. Here it is not, so the OU channel changes the relative likelihood of occupied and zero states. Define the one-coordinate Gaussian density $\varphi_\Delta(y) := (2\pi\Delta)^{-1/2}e^{-y^2/(2\Delta)}$. Summing over the three values of a_i gives

$$\begin{aligned} P_t(x_i \mid m) &= \varphi_{\Delta_t}(x_i)\frac{\sum_{a_i=-1,0,+1} e^{(u_m+\zeta_i)a_i - (\lambda+\kappa_t)a_i^2}}{D_\lambda(u_m)} \\ &= \varphi_{\Delta_t}(x_i)\frac{1 + 2e^{-(\lambda+\kappa_t)}\cosh(u_m+\zeta_i)}{1+2e^{-\lambda}\cosh u_m}. \end{aligned} \tag{62}$$

Conditional independence now yields the exact branch likelihood

$$\boxed{P_t(x \mid m) = \prod_{i=1}^{N}\varphi_{\Delta_t}(x_i)\prod_{i=1}^{N}\frac{1 + 2e^{-(\lambda+\kappa_t)}\cosh(u_m+\zeta_i)}{1+2e^{-\lambda}\cosh u_m}.} \tag{63}$$

Bayes' rule gives $P_t(m \mid x) \propto P_t(x \mid m)P_0(m)$. The factor $[D_\lambda(u_m)]^N$ from $P_0(m)$ cancels the denominators in (63); the common Gaussian factor cancels on normalization. Thus the exact latent posterior is

$$\boxed{P_t(m \mid x) = \frac{e^{-\beta N m^2/2}\displaystyle\prod_{i=1}^{N}[1 + 2e^{-(\lambda+\kappa_t)}\cosh(u_m+\zeta_i)]}{\displaystyle\int_{\mathbb{R}} \mathrm{d}v\, e^{-\beta N v^2/2}\prod_{i=1}^{N}[1 + 2e^{-(\lambda+\kappa_t)}\cosh(\beta v + h/N + \zeta_i)]}.} \tag{64}$$

This remains a one-dimensional integral over m, but its integrand generally depends on all of $x_1, \ldots, x_N$ separately. A scalar latent integration is not yet scalar *score closure*.

The same three-term sum gives the exact sitewise denoiser at fixed m:

$$\boxed{\mathbb{E}[a_i \mid x_i, m] = \frac{2e^{-(\lambda+\kappa_t)}\sinh(u_m+\zeta_i)}{1+2e^{-(\lambda+\kappa_t)}\cosh(u_m+\zeta_i)}.} \tag{65}$$

Averaging over (64) gives

$$\mathbb{E}[a_i \mid x] = \int_{\mathbb{R}} \mathrm{d}m\, \mathbb{E}[a_i \mid x_i, m] P_t(m \mid x), \tag{66}$$

and the universal exact OU identity yields

$$F_i(x,t) = -\frac{x_i}{\Delta_t} + \frac{e^{-t}}{\Delta_t}\mathbb{E}[a_i \mid x]. \tag{67}$$

This is a complete exact description, but it is not yet a closed function of the empirical mean.

B.4 High-noise expansion: why the branch evidence becomes scalar

We now use the three-delta approximation (59) and investigate times

$$t = \tfrac{1}{2}\log N + O(1), \qquad e^{-t} = O(N^{-1/2}), \quad \kappa_t = O(N^{-1}), \quad \zeta_i = O_{\mathbb{P}}(N^{-1/2}).$$

The conditional clean mean in branch α is $m_\alpha \in \{-m^*, 0, +m^*\}$. It is useful to introduce its occupancy and variance,

$$\begin{aligned} q_\alpha &:= \mathbb{E}[a_i^2 \mid m_\alpha] = \frac{2e^{-\lambda}\cosh(\beta m_\alpha)}{1+2e^{-\lambda}\cosh(\beta m_\alpha)}, \\ v_\alpha &:= \mathrm{Var}(a_i \mid m_\alpha) = q_\alpha - m_\alpha^2. \end{aligned} \tag{68}$$

Unlike in Ising Curie–Weiss, q_0 and $q_\pm$ need not be equal. We must therefore keep the quadratic likelihood terms instead of simply copying the two-branch likelihood-ratio expansion.

At fixed branch, divide its noisy likelihood by the common Gaussian baseline. The corresponding one-site factor is

$$L_\alpha(\zeta_i, \kappa_t) := \mathbb{E}\big[e^{\zeta_i a_i - \kappa_t a_i^2} \mid m_\alpha\big].$$

For bounded spins, Taylor expansion of its logarithm gives

$$\log L_\alpha(\zeta_i, \kappa_t) = m_\alpha \zeta_i - q_\alpha \kappa_t + \frac{v_\alpha}{2}\zeta_i^2 + O(|\zeta_i|^3 + \kappa_t|\zeta_i| + \kappa_t^2). \tag{69}$$

Summing over coordinates,

$$\log P_t(x \mid m_\alpha) = \sum_i \log \varphi_{\Delta_t}(x_i) + m_\alpha \sum_i \zeta_i - N q_\alpha \kappa_t + \frac{v_\alpha}{2}\sum_i \zeta_i^2 + o_{\mathbb{P}}(1). \tag{70}$$

At the chosen time scale, $N\kappa_t = O(1)$ and, under the noised law,

$$\frac{1}{N}\sum_i x_i^2 = \Delta_t + o_{\mathbb{P}}(1).$$

Consequently

$$\sum_i \zeta_i^2 = \frac{N e^{-2t}}{\Delta_t} + o_{\mathbb{P}}(1) = 2N\kappa_t + o_{\mathbb{P}}(1).$$

The occupation and variance terms therefore combine as

$$\begin{aligned} -N q_\alpha \kappa_t + \frac{v_\alpha}{2}\sum_i \zeta_i^2 &= -N\kappa_t(q_\alpha - v_\alpha) + o_{\mathbb{P}}(1) \\ &= -\frac{N e^{-2t}}{2\Delta_t} m_\alpha^2 + o_{\mathbb{P}}(1). \end{aligned} \tag{71}$$

The q_α-dependence cancels against the variance contribution. This cancellation is essential: the central and outer branches have different occupation statistics, yet no second collective observable is needed at leading order on this high-noise scale. We obtain

$$\boxed{\log P_t(x \mid m_\alpha) = C_t(x) + \frac{e^{-t}}{\Delta_t} m_\alpha \sum_i x_i - \frac{N e^{-2t}}{2\Delta_t} m_\alpha^2 + o_{\mathbb{P}}(1),} \tag{72}$$

where $C_t(x) = \sum_i \log \varphi_{\Delta_t}(x_i)$ is independent of the branch. The expansion is for typical noisy configurations; it is not an exact identity for every x.

Set

$$\mu := \frac{1}{\sqrt{N}} \sum_i x_i, \qquad c_t := m^* \sqrt{N} e^{-t}, \qquad \nu_t := \frac{c_t \mu}{\Delta_t}, \qquad d_t := \frac{c_t^2}{2\Delta_t}.$$

Combining (72) with the prior weights, the three branch posterior probabilities are

$$\boxed{\begin{aligned} p_0(\mu, t) &\simeq \frac{W_0}{D_t(\mu)}, \\ p_\pm(\mu, t) &\simeq \frac{W_\pm e^{\pm \nu_t - d_t}}{D_t(\mu)}, \\ D_t(\mu) &:= W_0 + e^{-d_t}[W_+ e^{\nu_t} + W_- e^{-\nu_t}]. \end{aligned}} \tag{73}$$

At $h = 0$, $W_+ = W_- = W$, so

$$\boxed{D_t(\mu) = W_0 + 2W e^{-d_t} \cosh(\nu_t).} \tag{74}$$

The factor e^{-d_t} is new relative to the symmetric *two*-branch calculation. It cannot be dropped: the central branch has squared mean zero while the outer branches have squared means $(m^*)^2$.

In each branch, (65) now gives $\mathbb{E}[a_i \mid x_i, m_\alpha] = m_\alpha + O(|\zeta_i| + \kappa_t)$. Averaging with (73),

$$\begin{aligned} \mathbb{E}[a_i \mid x] &\simeq -m^* p_-(\mu, t) + m^* p_+(\mu, t) \\ &= m^* \frac{2W e^{-d_t} \sinh(\nu_t)}{W_0 + 2W e^{-d_t} \cosh(\nu_t)} \qquad (h = 0). \end{aligned} \tag{75}$$

Thus the informative part of the full score becomes the same function of μ at every site, just as in binary Curie–Weiss, albeit with $\sinh / (W_0 + 2W \cosh)$ instead of a single tanh.

B.5 Projection and the candidate three-well potential

The increasing reverse clock is $s = t_f - t$. Project the full reverse OU SDE onto $\mu = N^{-1/2} \sum_i x_i$. The projected Brownian motion remains standard because

$$\frac{1}{N} \sum_{i,j} 2T \delta_{ij} \, \mathrm{d}s = 2T \, \mathrm{d}s.$$

Using (67) and the high-noise closed denoiser (75), the reverse drift becomes

$$\boxed{\begin{aligned} \mathrm{d}\mu_s &= b_3(\mu_s, t_f - s) \, \mathrm{d}s + \sqrt{2T} \, \mathrm{d}W_s^\mu, \\ b_3(\mu, t) &\simeq \left(1 - \frac{2T}{\Delta_t}\right) \mu + \frac{2T c_t}{\Delta_t} \frac{2W e^{-d_t} \sinh(\nu_t)}{W_0 + 2W e^{-d_t} \cosh(\nu_t)}. \end{aligned}} \tag{76}$$

This is now a scalar drift, and can be integrated. Since $\partial_\mu \nu_t = c_t / \Delta_t$, direct differentiation shows

$$\boxed{\begin{aligned} V_{3,\Delta}(\mu, t) = \frac{1}{2} \left(\frac{2T}{\Delta_t} - 1\right) \mu^2 \\ - 2T \log \left[W_0 + 2W e^{-c_t^2/(2\Delta_t)} \cosh\left(\frac{c_t \mu}{\Delta_t}\right)\right] + C(t), \end{aligned}} \tag{77}$$

with $b_3 = -\partial_\mu V_{3,\Delta}$ at the approximation order used. Only now, at large forward time, is it appropriate to set $\Delta_t \simeq T$. The particularly simple form is

$$\boxed{V_3(\mu,t) \simeq \frac{\mu^2}{2} - 2T\log\left[W_0 + 2We^{-c_t^2/(2T)}\cosh\left(\frac{c_t\mu}{T}\right)\right] + C(t).} \tag{78}$$

The additional constant-in-μ term *inside the logarithm* permits the central phase to compete with the two outer phases. Whether there are one or three visible wells depends on t, the phase separation, and the weights; three clean phases do not mean three reverse-time wells at every time.

B.6 Exact projected marginal: check and limitation of closure

There is an independent way to calculate a potential for μ, which is exact at every finite N. It also clarifies which part of the preceding argument is nontrivial. Unlike the auxiliary m, the *empirical* clean magnetization $S = \sum_i a_i$ takes integer values. At fixed $S = s$ let $q = Q = \sum_i a_i^2$ count occupied spins. Then

$$n_+ = \frac{q+s}{2}, \qquad n_- = \frac{q-s}{2}, \qquad n_0 = N - q.$$

The multinomial multiplicity and the occupancy cost give

$$C_N(s;\lambda) := \sum_{\substack{|s|\le q\le N \\ q\equiv s \pmod 2}} \frac{N!}{[(q+s)/2]![(q-s)/2]!(N-q)!}e^{-\lambda q}. \tag{79}$$

Hence the *exact* law of S is

$$\boxed{P_0(S=s) = \frac{C_N(s;\lambda)e^{\beta s^2/(2N)+hs/N}}{\displaystyle\sum_{r=-N}^{N} C_N(r;\lambda)e^{\beta r^2/(2N)+hr/N}}.} \tag{80}$$

Summing the forward OU coordinates gives, exactly,

$$\mu_t \mid S = s \sim \mathcal{N}\left(\frac{e^{-t}s}{\sqrt{N}}, \Delta_t\right). \tag{81}$$

Thus its one-dimensional density is the finite Gaussian sum

$$\boxed{P_t^\mu(\mu) = \sum_{s=-N}^{N} P_0(S=s)\,\varphi_{\Delta_t}\left(\mu - \frac{e^{-t}s}{\sqrt{N}}\right).} \tag{82}$$

Because μ_t is itself a forward OU process, its time reversal has an *exact* scalar potential

$$\boxed{V_{\text{marginal}}(\mu,t) = -\frac{\mu^2}{2} - 2T\log P_t^\mu(\mu) + C(t).} \tag{83}$$

This result does *not* require a three-phase approximation. At the speciation scale, concentration near m_α turns (82) into

$$P_t^\mu(\mu) \simeq W_-\varphi_{\Delta_t}(\mu + c_t) + W_0\varphi_{\Delta_t}(\mu) + W_+\varphi_{\Delta_t}(\mu - c_t),$$

which gives (77) by completing Gaussian squares. This is a useful independent check of the potential.

Marginal potential versus closure of the full score

For *any* clean prior, the linear coordinate $\mu = N^{-1/2}\sum_i x_i$ is itself an exact forward OU process, so its one-dimensional reversed marginal admits the potential (83). What is special about the present mean-field model is the stronger statement that the *projected full-state reverse drift*, evaluated at a microscopic configuration x, becomes a function of $\mu(x)$ alone at the high-noise scale. Equation (72) and the site-independent approximation in (75) are the ingredients that establish this latter, asymptotic closure.

Structural trend. The Ising case has two phases, while this three-state model has three coexisting phases when λ is tuned appropriately. Yet the high-noise branch evidence in both cases is controlled by the same scalar $\sum_i x_i$: the branches differ in their common mean along the uniform direction. The number of phases therefore need not equal the number of collective coordinates. The relevant requirement is that branch discrimination and the within-branch denoiser become functions of the proposed collective statistic, at the stated approximation order.

B.7 A first bridge toward n wells: number of phases versus latent dimension

The three-branch example suggests distinguishing the number of phases from the number of independent directions in which they differ. As a transparent reference case, consider K Gaussian branch laws with $X \mid B = \alpha \sim \mathcal{N}(v_\alpha, \Sigma)$, where the same positive-definite covariance Σ is used in each branch. Bayes' rule and expansion of the Gaussian exponents give, relative to branch 1,

$$\log \frac{P(B = \alpha \mid x)}{P(B = 1 \mid x)} = \log \frac{w_\alpha}{w_1} + x^\top \Sigma^{-1}(v_\alpha - v_1) - \frac{1}{2}\left(v_\alpha^\top \Sigma^{-1} v_\alpha - v_1^\top \Sigma^{-1} v_1\right).$$

Hence all posterior branch probabilities depend on x only through the linear projections onto the span of $\{\Sigma^{-1}(v_\alpha - v_1) : \alpha = 2, \dots, K\}$. The dimension of this span is at most $K - 1$, regardless of the ambient dimension. Several branches along one common direction can therefore be distinguished with just one scalar, whereas branches whose mean differences span several directions require several independent linear projections. This is a statement about *branch-posterior sufficiency*; full reverse-score closure requires the additional within-branch posterior means to depend only on those same projections, as shown for the two- and three-state mean-field models above. For non-Gaussian branch likelihoods this linear-sufficiency conclusion need not hold exactly.

For genuinely competing template fields $h^1, \dots, h^K$ on a common reference measure $\rho(\mathrm{d}a)$, the corresponding clean law is $P_0(\mathrm{d}a) = \sum_\alpha w_\alpha e^{\beta\langle a, h^\alpha\rangle} \rho(\mathrm{d}a)/Z_\alpha$. It must not be confused with multiplying all template factors:

$$e^{\beta \sum_\alpha \langle a, h^\alpha\rangle} = e^{\beta\langle a, \sum_\alpha h^\alpha\rangle},$$

which describes a single effective field, rather than a latent mixture. This elementary distinction underlies the outlook in the main text.

C Curie–Weiss reduction: complete derivation from the prior to the potential

Objective. This appendix presents the Curie–Weiss calculation as one continuous argument. Every approximation is introduced at the point where it first enters.

C.1 Clean prior, scaling, and Hubbard–Stratonovich decoupling

Start from

$$P_0(a) = \frac{1}{\mathcal{Z}_N} \exp\left[\frac{\beta}{2N}\left(\sum_i a_i\right)^2 + \frac{h}{N}\sum_i a_i\right].$$

If $m_a = N^{-1}\sum_i a_i$, then the interaction is $\beta N m_a^2/2$, so the $1/N$ factor makes it extensive. The field term is $\pm h m^*$ in the two ferromagnetic phases; it therefore creates a finite branch bias without suppressing one phase exponentially in N.

Let $S = \sum_i a_i$. Completing the square gives

$$-\frac{\beta N}{2}m^2 + \beta m S = -\frac{\beta N}{2}\left(m - \frac{S}{N}\right)^2 + \frac{\beta S^2}{2N}. \tag{84}$$

Using the Gaussian integral,

$$\exp\left(\frac{\beta S^2}{2N}\right) = \sqrt{\frac{\beta N}{2\pi}}\int_{\mathbb{R}} \mathrm{d}m\, \exp\left[-\frac{\beta N}{2}m^2 + \beta m S\right]. \tag{85}$$

Substitution into the clean law gives the exact augmented joint measure

$$P_0(m, a) = \frac{1}{\widetilde{\mathcal{Z}}_N}\exp\left(-\frac{\beta N}{2}m^2\right)\prod_{i=1}^N \exp[(\beta m + h/N)a_i]\,. \tag{86}$$

Integrating over m recovers the original $P_0(a)$. At fixed m, factors that depend only on m cancel from the conditional distribution, so

$$P_0(a \mid m) = \prod_{i=1}^N \frac{e^{(\beta m + h/N)a_i}}{2\cosh(\beta m + h/N)}. \tag{87}$$

Thus the interacting measure becomes a mixture of product measures.

C.2 Marginal magnetization law, phase concentration, and branch masses

Summing (86) over all spin configurations gives

$$\begin{aligned} P_0(m) &\propto \exp\left(-\frac{\beta N}{2}m^2\right)[2\cosh(\beta m + h/N)]^N \\ &\asymp \exp[-\beta N \Phi_N(m)], \end{aligned} \tag{88}$$

where, up to an additive constant independent of m,

$$\Phi_N(m) = \frac{m^2}{2} - \frac{1}{\beta}\log\cosh(\beta m + h/N).$$

Stationary points satisfy

$$m = \tanh(\beta m + h/N).$$

For $\beta > 1$ and large N, there are stable minima $m_\pm = \pm m^* + O(N^{-1})$, with $m^* = \tanh(\beta m^*)$.

The mass near either minimum is obtained by Laplace's method:

$$I_\pm := \int_{\text{near } m_\pm} e^{-\beta N \Phi_N(m)}\, \mathrm{d}m \simeq e^{-\beta N \Phi_N(m_\pm)} \sqrt{\frac{2\pi}{\beta N \Phi_N''(m_\pm)}}. \tag{89}$$

The curvature controls the peak width $O(N^{-1/2})$, while the exponential factor controls its mass. Expanding the weak field,

$$\Phi_N(m) = \Phi_0(m) - \frac{h}{\beta N}\tanh(\beta m) + O(N^{-2}),$$

so

$$e^{-\beta N \Phi_N(m_\pm)} \simeq e^{-\beta N \Phi_0(m^*)} e^{\pm h m^*}.$$

Since the two curvature factors agree at leading order,

$$I_+ : I_- = e^{h m^*} : e^{-h m^*},$$

and normalization gives (13). Therefore

$$P_0(m) \simeq W_+ \delta(m - m^*) + W_- \delta(m + m^*). \tag{90}$$

Conditional on branch $\pm$,

$$P_0(a \mid \pm) = \prod_{i=1}^N \frac{e^{\pm \beta m^* a_i}}{2\cosh(\beta m^*)}, \qquad \mathbb{E}[a_i \mid \pm] = \pm m^*. \tag{91}$$

C.3 Exact branch algebra after OU diffusion

For a single binary spin,

$$(x_i - e^{-t} a_i)^2 = x_i^2 + e^{-2t} - 2 x_i e^{-t} a_i, \tag{92}$$

so the spin-dependent part of the OU likelihood is $\exp(a_i \zeta_i)$ with

$$\zeta_i = \frac{x_i e^{-t}}{\Delta_t}.$$

Multiplying by the branch prior gives the local posterior field

$$A_i^\pm = \pm \beta m^* + \zeta_i.$$

The binary identity

$$\frac{\sum_{a=\pm 1} a e^{aA}}{\sum_{a=\pm 1} e^{aA}} = \tanh A$$

therefore yields

$$\mathbb{E}[a_i \mid x, \pm] = \tanh(\pm \beta m^* + \zeta_i). \tag{93}$$

Summing the clean spins inside each branch gives

$$\frac{\cosh(\pm \beta m^* + \zeta_j)}{\cosh(\beta m^*)} = \cosh \zeta_j \, [1 \pm m^* \tanh \zeta_j].$$

The $\cosh \zeta_j$ factors are common to both branches. Their branch-dependent likelihood factors are therefore exactly

$$Q_\pm(x) = \prod_j [1 \pm m^* \tanh \zeta_j],$$

and Bayes gives (15). The total posterior mean is

$$\mathbb{E}[a_i \mid x] = \frac{W_+ Q_+(x) \tanh(\beta m^* + \zeta_i) + W_- Q_-(x) \tanh(-\beta m^* + \zeta_i)}{W_+ Q_+(x) + W_- Q_-(x)}. \tag{94}$$

This is exact within the two-delta-peak thermodynamic model.

C.4 Controlled large-time closure

Assume $\max_i |\zeta_i| \ll 1$. Instead of separately approximating Q_+ and Q_-, consider their ratio:

$$\log \frac{Q_+}{Q_-} = 2 \sum_j \operatorname{artanh}(m^* \tanh \zeta_j)$$
$$= 2m^* \sum_j \zeta_j + O\left(\sum_j |\zeta_j|^3 \right). \tag{95}$$

Using $W_+/W_- = e^{2hm^*}$,

$$\log \frac{P(+ \mid x)}{P(- \mid x)} = 2m^* \left(h + NM(x) \frac{e^{-t}}{\Delta_t} \right) + O\left(\sum_j |\zeta_j|^3 \right). \tag{96}$$

Define

$$A(x,t) := m^* \left(h + NM(x) \frac{e^{-t}}{\Delta_t} \right).$$

Then

$$P(+ \mid x) - P(- \mid x) \simeq \tanh A(x,t).$$

At the same order,

$$\mathbb{E}[a_i \mid x, \pm] = \pm m^* + O(\zeta_i).$$

Substitution into (94) gives

$$\mathbb{E}[a_i \mid x] \simeq m^* \tanh A(x,t),$$

which is (17). It is independent of i and depends on x only through $M(x)$. The exact OU score identity now gives (18).

At the transition scale $t = \frac{1}{2} \log N + O(1)$, typical high-noise coordinates satisfy

$$\max_i |\zeta_i| = O_{\mathbb{P}}\left(\sqrt{\frac{\log N}{N}} \right),$$

so the coordinatewise expansion is self-consistent. The cubic remainder in (95) is then controlled under the same typical-event estimate.

C.5 Projection, exact-in-Δ_t potential, and large-time form

Define $\mu = N^{-1/2} \sum_i x_i$. The projected noise has covariance

$$\langle \eta_\mu(t) \eta_\mu(t') \rangle = \frac{1}{N} \sum_{i,j} 2T \delta_{ij} \delta(t - t') = 2T \delta(t - t'). \tag{97}$$

Summing the closed score gives

$$\frac{1}{\sqrt{N}} \sum_i F_i^{\mathrm{LT}}(x,t) = -\frac{\mu}{\Delta_t} + m^* \sqrt{N} \frac{e^{-t}}{\Delta_t} \tanh \left[m^* \left(h + \sqrt{N} \mu \frac{e^{-t}}{\Delta_t} \right) \right].$$

Substitution in the reverse equation gives (21). Direct differentiation verifies

$$-\partial_\mu V_\Delta = \left(1 - \frac{2T}{\Delta_t} \right) \mu + 2Tm^* \sqrt{N} \frac{e^{-t}}{\Delta_t} \tanh \left[m^* \left(h + \frac{\sqrt{N} e^{-t}}{\Delta_t} \mu \right) \right].$$

Thus (22) is the effective potential before the approximation $\Delta_t \simeq T$.

At large time, $\Delta_t = T(1 - e^{-2t}) = T[1 + o(1)]$, and the potential reduces to (26). In the increasing reverse clock, its minima are stable fixed points. For $h = 0$, the curvature at the origin is

$$V''(0,t) = 1 - \frac{2m^{*2} N}{T} e^{-2t},$$

which changes sign at (30). For smaller t, $\log \cosh Z \sim |Z|$ gives the two-well form HO IL PENE ENORME (31).